

Module 2 Summary



Module 2 Review, Reflect, Consolidate, and Practice

Module 2 built directly on Module 1. In Module 1, you learned how to compute returns and visualise financial data. In this module, you learned how to interpret uncertainty, measure it rigorously, and connect it to real-world financial decisions.

This section summarises what you have learned and links it explicitly to the course learning outcomes.

What You Learned in Module 2

1. Risk Is Uncertainty About Returns

You learned that:

- Return tells you how much you gain or lose.
- Risk tells you how uncertain that gain or loss is.
- Two investments can have the same average return but very different risk profiles.

You now understand that risk is not simply about losses. It is about **variability in outcomes**, which can be measured quantitatively.

This directly supports:

CLO1: Explain key concepts and theoretical foundations of financial analytics and their role in strategic decision-making.

You can now clearly explain what financial risk is and why it matters.

2. Time Series Properties Matter for Risk Measurement

You learned that:

- Prices are typically non-stationary.
- Returns are approximately stationary.
- Volatility evolves over time.
- Volatility clustering means high-volatility periods follow high-volatility periods.

You saw that:

- Risk must be measured on returns, not prices.
- Variance and standard deviation are meaningful only when statistical properties are stable.
- Financial markets exhibit persistence in volatility.

You visualised:

- Daily return series
- Periods of calm and turbulence
- Clusters of volatility
- Autocorrelation in squared returns

This supports:

CLO2: Apply quantitative and time-series techniques to model risk, return, and financial performance in real-world contexts.

You have now applied time-series reasoning directly to financial data.

3. Volatility Is a Dynamic Measure of Risk

You learned that volatility:

- Is measured using standard deviation of returns.
- Changes over time.
- Can be estimated dynamically using rolling windows.
- Can be measured using realised volatility from daily returns within each month.

You implemented:

- Monthly volatility from daily returns
- Rolling volatility estimation
- Visual diagnostics of clustering

You now understand that:

Risk today depends on recent market behaviour.

This is foundational for portfolio management, derivatives pricing, and regulatory capital decisions.

4. Downside Risk: VaR and Expected Shortfall

You learned to move beyond standard deviation and measure **tail risk**.

You defined:

Value at Risk (VaR)

You learnt about and interpreted:

- VaR in return terms
- VaR in dollar terms
- The limitations of VaR

You then extended this to:

Expected Shortfall (ES)

Expected Shortfall measures:

The average loss conditional on being in the worst 5% of outcomes.

You saw why ES provides more information than VaR, especially during crises.

This supports:

CLO2: Apply quantitative and time-series techniques to model risk, return, and financial performance in real-world contexts. and

CLO3: Use business analytics tools to develop, evaluate, and communicate predictive insights that support financial problem-solving.

You applied quantitative techniques and interpreted them in financial decision contexts.

5. Communicating Risk Visually

You created:

- Return histograms
- VaR threshold overlays
- Volatility time series
- Comparative volatility plots

For example:

```
ggplot(aapl_daily_ret, aes(x = daily.returns)) +  
  geom_histogram(bins = 100) +  
  geom_vline(xintercept = aapl_var_95, colour = "red") +  
  labs(  
    title = "Apple Daily Returns with 95% VaR",  
    x = "Daily Return",
```

```
y = "Frequency"
```

```
) +
```

```
theme_bw()
```

You now understand how to:

- Show the left tail of the distribution.
- Communicate risk thresholds visually.
- Explain probabilistic risk clearly.

This supports:

CLO4: Create and communicate financial insights using effective visual, written, and oral formats suited to both technical and non-technical stakeholders.

You can now explain risk in both statistical and business language.

Practice Tasks

Note: For the first three tasks, utilize the R scripts provided throughout the module. The R script for the fourth task is provided.

Practice Task 1: Comparing Volatility

1. Download daily data since January 1, 2010, for:
 - ■ Pfizer Inc. (NYSE: PFE)
 - Johnson & Johnson (NYSE: JNJ)
2. Compute daily returns.
3. Calculate:
 - Mean daily return
 - Standard deviation of daily returns
4. Interpret:
 - Which asset is riskier?
 - Does higher volatility correspond to higher average return?

Write 5 to 6 sentences explaining your findings.

Practice Task 2: Visualising Clustering

1. Plot daily returns for Tesla.
2. Identify visually:
 - One calm period
 - One turbulent period
3. Compute squared returns and plot them.

4. Write:

- Do you observe clustering?
- What does this imply for risk measurement?

Practice Task 3: VaR and Expected Shortfall

1. Compute 95% daily VaR.
2. Compute 95% Expected Shortfall.
3. Convert both to dollar values for a \$5 million portfolio.
4. Explain:
 - What does VaR tell you?
 - What does ES tell you that VaR does not?

Practice Task 4: Rolling Volatility [Hint: Use the `tq_mutate` function and set `mutate_fun = rollapply`]

1. Compute 30-day rolling volatility for one stock.
2. Plot the rolling volatility series.
3. Identify:
 - A spike
 - A sustained calm period
4. In 4 to 6 sentences, explain what this means for portfolio risk management.

Example solution for Task 4 available [here. \(https://learn.usc.edu.au/courses/35359/files/3292571?wrap=1\)](https://learn.usc.edu.au/courses/35359/files/3292571?wrap=1)  [\(https://learn.usc.edu.au/courses/35359/files/3292571/download?download_frd=1\)](https://learn.usc.edu.au/courses/35359/files/3292571/download?download_frd=1)

Reflections

Write short responses to the following:

1. Why is it misleading to measure risk using price variability instead of return variability?
2. How does volatility clustering change how a bank should think about capital requirements?
3. If volatility suddenly increases, what strategic decisions might a portfolio manager consider?
4. What are the limitations of VaR as a risk measure?

Workshop

[Tasks \(https://learn.usc.edu.au/courses/35359/files/3282924?wrap=1\)](https://learn.usc.edu.au/courses/35359/files/3282924?wrap=1)